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mysql> CREATE TABLE Students (
->   roll_no INT PRIMARY KEY,
->   name VARCHAR(50) NOT NULL,
->   email VARCHAR(100) UNIQUE,
->   age INT CHECK (age >= 18),
->   city VARCHAR(50) DEFAULT 'Nagpur'
-> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> INSERT INTO Students (roll_no, name, email, age, city) VALUES
->   (1, 'Alice', 'alice@mail.com', 20, 'Mumbai'),
->   (2, 'Bob', 'bob@mail.com', 22, 'Nagpur'),
->   (3, 'Charlie', 'charlie@mail.com', 19, 'Pune');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> CREATE TABLE Results (
->   result_id INT PRIMARY KEY,
->   roll_no INT,
->   marks INT,
->   FOREIGN KEY (roll_no) REFERENCES Students(roll_no) ON DELETE CASCADE ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> INSERT INTO Results VALUES
->   (101, 1, 85),
->   (102, 2, 90),
->   (103, 3, 78);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+-----+
| roll_no | name  | email          | age | city  |
+-----+-----+-----+-----+-----+
| 1       | Alice | alice@mail.com | 20  | Mumbai |
| 2       | Bob   | bob@mail.com   | 22  | Nagpur |
| 3       | Charlie | charlie@mail.com | 19  | Pune   |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Results;
+-----+-----+-----+
| result_id | roll_no | marks |
+-----+-----+-----+
| 101       | 1       | 85    |
| 102       | 2       | 90    |
| 103       | 3       | 78    |
+-----+-----+-----+
3 rows in set (0.00 sec)

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mysql>
mysql> -- Domain violation: age < 18
mysql> INSERT INTO Students VALUES (4, 'Sam', 'sam@mail.com', 15, 'Delhi');
ERROR 3819 (HY000): Check constraint 'students_chk_1' is violated.
mysql>
mysql> -- Integrity violation: duplicate roll_no
mysql> INSERT INTO Students VALUES (2, 'Eve', 'eve@mail.com', 21, 'Delhi');
ERROR 1062 (23000): Duplicate entry '2' for key 'students.PRIMARY'
mysql>
mysql> -- Referential violation: roll_no not in Students
mysql> INSERT INTO Results VALUES (104, 10, 75);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
mysql>
mysql> DELETE FROM Students WHERE roll_no = 3;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+-----+
| roll_no | name  | email          | age | city  |
+-----+-----+-----+-----+-----+
| 1       | Alice | alice@mail.com | 20  | Mumbai |
| 2       | Bob   | bob@mail.com   | 22  | Nagpur |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Results;
+-----+-----+-----+
| result_id | roll_no | marks |
+-----+-----+-----+
| 101       | 1       | 85    |
| 102       | 2       | 90    |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> DROP DATABASE constraints_demo;
Query OK, 2 rows affected (0.03 sec)

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